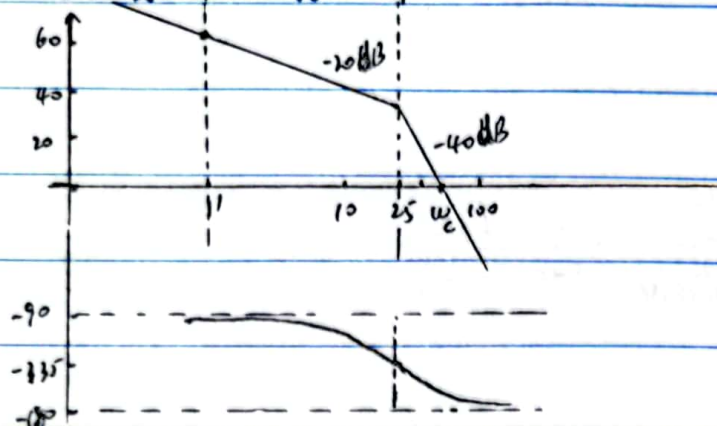


$$1. \lim_{s \rightarrow 0} s \cdot \frac{1}{s^2} \frac{1}{1+G_0(s)} = \lim_{s \rightarrow 0} \frac{1}{s} \frac{1}{1+\frac{K}{s(s+25)}} = \frac{1}{K_v} \Rightarrow K = 25K_v = 1250$$

绘制波特图如下 未按比例



$$20 \lg |G(j\omega_c)| = 0 \Rightarrow \omega_c = 31.24$$

代入 ω_c

$$G(j\omega_c) = \frac{K}{j\omega_c(j\omega_c+25)}$$

$$\varphi(\omega_c) = -141.332^\circ$$

$$\therefore \gamma_c = -141.332^\circ + 180^\circ = 38.668^\circ$$

$$\therefore \gamma_{pm} = 55^\circ - \gamma_c + 5^\circ = 21.332^\circ$$

$$\therefore a = \frac{1 + \sin \gamma_{pm}}{1 - \sin \gamma_{pm}} = 2.1435$$

$$L(\omega_c') = -20 \lg \sqrt{a}$$

$$\therefore 20 \lg \left| \frac{K}{j\omega_c(j\omega_c+25)} \right| = -20 \lg \sqrt{a} \Rightarrow \left| \frac{K}{j\omega_c(j\omega_c+25)} \right| = \frac{1}{\sqrt{a}} \Rightarrow \omega_c' = 39.285$$

$$\therefore T = \frac{1}{\omega_c' \sqrt{a}} = 0.01738 \quad \therefore \omega_1 = 57.530 \quad \omega_2 = 26.839$$

$$\therefore G(s) = 2.144 \cdot \left(\frac{1}{2.144} \frac{0.03725s+1}{0.01738s+1} \right) \frac{1250}{s(s+25)}$$

$$\text{验证: } \omega_c \approx 41, \quad \gamma = 180^\circ + \angle G(j\omega_c) = 180^\circ - 121.31^\circ = 58.69^\circ > 55^\circ$$

$$2. G_0(s) = \frac{7}{\frac{1}{12}s^3 + \frac{2}{3}s^2 + s} \quad \therefore G_0(j\omega) = \frac{7}{-\frac{1}{12}j\omega^3 + \frac{2}{3}\omega^2 + j\omega} \Rightarrow \omega_c = 3.2376 \text{ rad/s}$$

$$\varphi(\omega_c) = -176.65^\circ \quad \therefore \gamma = 3.35^\circ \quad \Delta\gamma = 40^\circ + 10^\circ = 50^\circ$$

$$\therefore \varphi(\omega_c') = -130^\circ \Rightarrow \omega_c' \approx 1.128 \quad \therefore |G(j\omega_c')| = 5.3122$$

$$\therefore -20 \lg |G(j\omega_c')| = 20 \lg b \Rightarrow b = 5.3122^{-1} = 0.18825$$

$$\therefore \frac{1}{bT} = \frac{1}{5} \omega_c' \Rightarrow T = \frac{5}{b\omega_c'} = 23.55 \quad \therefore \omega_1 = 0.0425 \quad \omega_2 = 0.2256$$

$$\therefore G(s) = \frac{4.4338+1}{23.55s+1} \frac{7}{s(0.55s+1)(\frac{1}{6}s+1)}$$

$$\text{验证: } K_v = 7 \quad \gamma = 180^\circ + \angle G(j\omega_c) = 40.5^\circ$$

